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30 June 1982
SUPERSEDING
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MILITARY SPECIFICATION

TRUCK, FIRE FIGHTING: STRUCTURAL, 1000 GPM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial 1000 gallon per minute (gpm) fire truck for use in combating structural fires at military installations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-320	- Chromium Plating (Electro-deposited).
QQ-P-35	- Passivation Treatment for Corrossion Resisting Steel.
VV-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-F-24385	- Fire Extinguishing Agent, Aqueous Film Forming Foam (AFFF) Liquid Concentrate, Six Percent, for Fresh and Sea Water.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Service.
FED-STD-297	- Rust Proofing of Automotive Vehicles.
FED-STD-595	- Colors.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Material.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORATION (DOT)

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNDERWRITERS LABORATORIES, INC. (UL)

UL Standards 822 - Guide for Certification of Fire Department Pumpers.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 1901.
NFPA No. 1931.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION (TRA)

Tire and Rim Association Yearbook, with Military Supplement.

(Application for copies should be addressed to the Tire and Rim Association, 3200 West Market Street, Akron, OH 44313.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Sheet Plate, Steel and Strip for Fusion-Welded Unfired Pressure Vessels; Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The fire truck (hereinafter called "truck") shall be a commercial, 4 x 2 front-wheel-steer, rear-wheel-drive, truck cab and chassis powered by a diesel engine. The truck shall be equipped with a midship-mounted pump driven by the truck engine, a water tank, and such other equipment and accessories as are specified herein. Operating parts not specifically covered herein but necessary to provide a complete and efficient truck shall be furnished.

3.2 First article. The contractor shall furnish one or more trucks as specified (see 6.2), for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his pro- duction methods and choice of design detail will produce trucks that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more trucks (see 6.2), for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9 and 6.10). Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other unless the metals are protected against galvanic corrosion.

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3.5 Enviromental operation requirements.

3.5.1 Temperature. The truck shall be capable of starting and performing in any temperature from plus 125° F to minus 25° F.

3.5.2 Altitude. The truck shall deliver its rated pumping capacity at 150 psi net pump pressure at any elevation from sea level to 2000 feet at the temperatures specified in 3.5.1. Net pump pressure shall be calculated as established by NFPA No. 1901-63. When specified (see 6.2), modification kits shall be provided and installed for utilization of trucks at specified elevations above 2000 feet.

3.6 Safety. All exposed parts which are subject to high operating temperatures, or which are energized electrically shall be insulated, fully enclosed or guarded. All rotating, reciprocating or moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Fire fighting equipment, when mounted on the truck, shall be located and positioned so as not to present a safety hazard to personnel. Surfaces of the truck which are to be used as walkways, working areas, or steps, shall be provided with non-skid surfaces.

3.7 Human factors. The characteristics of the truck shall provide for operation by personnel ranging from the small man, clothed, through the large man, arctic-coated, in accordance with SAE J833 and J925. Universal symbols in accordance with SAE J1500 shall be used throughout.

3.7.1 Sound level.

3.7.1.1 Exterior noise. The exterior sound level of the truck shall not exceed 83dB(A) when measured in accordance with SAE J366.

3.7.1.2 Personnel noise. The sound level at the operators position shall not exceed 85dB(A) as specified for category D of MIL-STD-1474. It shall be tested in accordance with 4.5.2.4.2.

3.8 Ease of maintenance. All major assemblies, including pumps, transmission, hose reels, foam proportioner, water and foam tanks, and similar items, shall be arranged for individual removal and replacement without major disassembly or removal of other components of the truck body. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility.

3.9 Performance. The truck, loaded as specified herein and with filled water, foam, fuel, and lubricant compartments, a 1500 pound evenly distributed simulated load, and with either a crew of five men, weighing not less than 175 pounds each, or an equivalent deadweight, shall be capable of meeting all performance and mobility requirements of NFPA 1901.

3.10 Truck chassis. The chassis shall be cab-ahead-of-engine type (custom cab).

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3.10.1 Gross vehicle weight rating. The truck manufacturer's gross vehicle weight (gvw) rating for the truck chassis shall be not less than 27,000 pounds. The total gross weight of the truck when loaded as specified in 3.9 shall not exceed the published rating of the axle manufacturer and the tire manufacturer.

3.10.2 Wheel base. The wheel base shall be of such length so as to mount the body and the equipment specified herein.

3.10.2.1 Dimensions. The overall dimensions of the truck including all lights and mounted equipment shall not exceed the following:

Height	115 inches
Length	336 inches
Width	96 inches (not including mirrors)

When a water tower is specified (see 3.21 and 6.2).

Height	132 inches
Length	384 inches
Width	96 inches (not including mirrors)

3.10.3 Frame. The chassis frame shall be of a heavy-duty design for fire service use and shall be provided with crossmembers, exclusive of engine supports, to support the gross weight of the body and load, the powerplant, the pump, the filled water tank, and all other equipment under the operating conditions specified herein, without evidence of distortion adversely affecting operation. No alterations shall be made to the frame which will reduce its strength.

3.10.4 Engine. The engine shall be the manufacturer's current standard heavy-duty liquid-cooled diesel type. The engine shall have not less than six cylinders, with replaceable cylinder liners. The engine shall develop not less than 280 brake horsepower (bhp) at a speed of not more than 2000 revolutions per minute (rpm) when operating on diesel fuel conforming to VV-F-800 and lube oil conforming to MIL-L-2104 and MIL-L-46167. The power and speed rating of the engine shall be such that when the truck is operated as specified herein it shall not require more than the EPA (Environmental Protection Agency) certified horsepower for that engine. Only full-flow type oil filters shall be utilized in the engine lubricating system. An engine governor shall be provided and set to limit engine speed as recommended by the engine manufacturer. The engine starting system shall be furnished, or approved, by the engine manufacturer. The installation of the engine in the truck shall be such that the engine crankshaft can be manually turned or barred. The engine shall be furnished with an engine governor set and sealed by the engine manufacturer to limit the maximum speed of the engine.

3.10.4.1 Engine fuel system. The fuel system shall be standard for the vehicle furnished. All components of the fuel system shall be located or shielded so as not to be subject to vapor lock. Fuel lines shall be adequately

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supported and shall be protected against mechanical damage. Construction and mounting shall be such that neither vibration nor tightening of mounting devices will cause leaks to develop. The bottom of the fuel tank and its supporting members shall be not lower than the lowest adjacent parts of the chassis or body. The tank capacity shall be large enough to operate the truck for not less than 4 hours at rated output without refueling. The tank shall hold not less than 50 gallons. Construction of the filler opening shall permit filling from the spout of a military 5-gallon liquid container and shall be located so that spills or overflow cannot run down on the engine exhaust system or batteries. The fuel tank shall be located on the truck so as not to be affected by heat from the exhaust system. The filler cap shall not protrude past the side of the truck.

3.10.4.2 Engine cooling system. The cooling system shall be standard for the truck furnished. When required (see 6.2) a supplemental heat-exchanger cooling system shall be provided which permits delivery of water from the fire pump to the heat-exchanger for cooling engine coolant without admixture. The supplemental heat-exchanger shall be supported with brackets. Draincocks shall be installed at the low points of the cooling system and at such points as may be necessary to drain the system to prevent freezing; the cocks shall be of a hand-operated type that will remain closed under the most severe operating conditions to be encountered by the truck. A coolant temperature indicator shall be provided on the cab dash panel and also on the pump control panel. The heat-exchanger cooling water control shall be located on the pump control panel.

3.10.4.3 Engine lubrication. The engine lubrication system shall be the engine manufacturers standard practice and shall include an oil filter.

3.10.4.4 Exhaust system. The truck shall be equipped with a muffler and exhaust system. The tailpipe shall be sized for the engine used and shall be located in such a manner that entrance of exhaust gases into the cab shall be minimized under all conditions of operation. Location of the tailpipe shall prevent heating of tires or the potential for heat from the tailpipe creating a fire hazard. The muffler, exhaust pipe, and tailpipe shall be of high-grade materials for the purpose used.

3.10.4.5 Brake horsepower certification. The contractor shall provide a certified net brake horsepower curve for the engine being furnished. The contractor shall further certify that the engine furnished complies with the requirements of this specification.

3.10.5 Electrical equipment. The truck shall be equipped with a 12-volt electrical system. All electrical components shall be designed so that their normal functioning will not be impaired by entrance of water during heavy rains, from road splash, condensation, or water spray resulting from other causes. Wiring shall be color coded or numbered for easy identification and secured (except for ignition harness) to the frame and body throughtout the truck with fasteners. Means of wiring identification shall be consistent throughout the

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truck. Wire shall be protected by means of looms or grommets at points of potential abrasion. Wiring shall be neatly run with all excess wire removed. All components shall be designed for operation at 12 volts. The use of 6-volt equipment paired in series with resistors or tapped to the midpoint of the battery is not acceptable. The truck body electrical circuits, excluding truck chassis electrical circuits, shall be provided with overload protective devices of the automatic reset, overload-circuit-breaker type.

3.10.5.1 Starting system. The starting system shall be standard for the type truck furnished. The ignition system shall be furnished with a keyless switch and a pilot light. Trucks shall be provided with a safety device to prevent engine starting unless the transmission is in a neutral position.

3.10.5.2 Battery units and electrical circuits. Two 12-volt battery units shall be provided. Both battery units shall be housed and shall be mounted on either a draw-out-type or permanent-type tray to permit accessibility for testing and servicing. The permanent-type tray may be mounted on the running board of the truck, provided it does not protrude beyond the sides of the truck. Provisions shall be made for locking draw-out-type tray in a retracted position, and battery cables shall be arranged to prevent chafing. Batteries shall be securely held in place by means of removable locking devices and shall be protected against engine and exhaust heat. Battery compartments shall be ventilated to the truck exterior and treated against acid fumes and spillage. The batteries shall be shipped charged and dry. Each battery shall be provided with a polarized receptacle for station charging and shall be located so as to automatically be disconnected when the truck moves forward. The battery circuit shall be provided with a reverse current device to protect the voltage regulator from damage when the batteries are connected in reverse polarity. The receptacle shall be of the recessed male type with a spring-loaded cover, and shall be provided with a matching female plug. The batteries shall conform to SAE Number 20T6A or 20T8A. Battery circuits and switching systems shall be provided whereby:

(a) Either battery may be used:

  (1) For ignition and lights alone.
  (2) For starting, alone.
  (3) For starting, ignition and lighting.

(b) Both batteries may be used simultaneously in starting.
(c) Neither the lighting wiring nor the ignition wiring will be connected to both batteries except when both starter switches are operated.
(d) The alternator shall be connected to supply the battery which is being used for ignition and lights.
(e) Wiring shall conform to wiring diagram entitled "Suggested Application of Dual Batteries to Single Ignition Engine" in the appendix of NFPA No. 1901.
(f) Batteries shall be sufficient to start engine after truck has been cold soaked for 24 hours at minus 25° F.

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3.10.5.2.1 Electric generating system. The engine shall be equipped with a 12 volt alternator and regulator capable of supplying the built-in DC current requirements of the truck with a minimum rating of 145 amps. Charging rate at 1000 engine rpm shall be minimum of 70 amps. A 6000 watt AC power inverter capable of continuous operation at rated output shall be provided. The inverter shall electrically attach to the vehicle alternator and shall produce constant 60 Hz plus or minus 1 Hz power. The inverter shall have voltage control of plus or minus 5 percent no-load to full load. Pulley ratio driving the alternator shall be sized to provide a minimum of 2000 watts of AC power at the engine manufacturer's rated idle speed of the engine. A remote control shall be installed to allow starting and stopping of the inverter from the pump operator's position and the driver's cab position. A voltmeter shall be provided with both control positions to indicate inverter operation. All wires shall be of sufficient gage and shall be properly supported and protected from injury. Four fused, grounded, weatherproof, dual receptacles, with spring-loaded covers, shall be internally wired into the 120 volt AC output of the inverter. An outlet shall be mounted on each side of the truck adjacent to the pump compartment protected from water spray and two outlets shall be located for connecting the lights specified in 3.25(j). The two outlets for the lights shall be provided with on-off switches mounted on the cab dash. The inverter shall be mounted in the floor of the cab between the driver and the passenger seats.

3.10.5.3 Lighting system. All truck lights, reflectors, and wiring shall conform to DOT Motor Carrier Safety Regulations, except as specified herein:

(a) DOT identification lamps are required.
(b) Clearance and side marker lamps shall be combined and mounted to indicate width of trucks.
(c) Overload protective devices shall be of the automatic-reset, overload-circuit-breaker type. The lighting system shall be arranged so that the headlamp circuit or circuits shall not be affected by a short circuit in any of the other circuits.
(d) Rear-mounted lights shall be located so that they shall not be obscured by equipment or personnel normally carried on the rear step.
(e) Front-turn-signal units shall be single-faced flush mounted. The self-canceling-type controller shall be mounted on the steering column, shall operate a visual indicator, and shall be wired through the ignition system.

3.10.5.3.1 Additional lighting. The following additional lighting shall be provided:

(a) Two automotive-type sealed-beam spotlights located on each side of the windshield, with controls extended inside the cab for "on-off" control and movement of the lamp housing through horizontal and vertical angles. Spotlight control handle shall not obscure the driver's vision.

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(b) Two sealed-beam hose pickup lights. Lights shall be mounted to the outside of the hose bed.
(c) Two shielded engine-compartment lights arranged to illuminate both sides of the engine. Both lights shall be controlled by one switch installed in the cab.
(d) Pump control panel lights enclosed within hoods or covers designed to completely illuminate the panel but prevent glare.
(e) Two backup lights, with not less than 5-inch diameter, located at rear of the body. The backup light switch shall be automatically controlled when the transmission is placed in the reverse mode.
(f) Stoplights, taillights, rear-turn signal lights, and backup lights shall be flush mounted at the rear of the body.
(g) A complete wiring diagram shall be provided with each manual furnished with the truck.
(h) Step lights for the cab and jump seat; controlled by the headlight switch. Lights shall be hooded or shielded for downward illumination.

3.10.5.3.2 Warning lights. Lighting shall conform to NFPA 1901 to include, not less than 3 sealed-beam lights and a chromium-plated base shall be centric mounted on top of the truck cab. When a water tower is specified, two lights shall be provided, one centric roof mounted on either side of the tower boom. Two motorized oscillating sine-wave-pattern lights shall be mounted equally spaced on the front of the cab under the windshield. All lights shall have red lenses. Two strobe type warning red lights shall be mounted, one on either side at the rear of the truck outside of the hose pickup lights. Lights shall be mounted so as not to interfere with packing or laying hose. The controls for the warning lights shall be independent of all other electrical equipment and shall be broken down into three groups: front, top, and rear. All lights shall be 12 volts DC.

3.10.5.3.3 Light switches. The headlight switch shall be the standard push-pull type. The warning light shall be the single-pole-single-throw, rocker, circuit-breaker type with indicating light for the "on" position. There shall be one switch for each of the groups specified in 3.10.5.3.2. All switches shall be clearly and permanently marked.

3.10.5.4 Siren and horn. The truck shall be equipped with a Federal Sign and Signal Corporation electronic siren and PA combination Model PA20A, or equal. The electronic siren shall be furnished with a microphone and all wiring applicable to two-way radio. A single 6-inch-diameter chromium-plated speaker shall be mounted on a chromium-plated bracket between the two sine wave lights. A Federal Sign and Signal Corporation 12-volt siren Model Q2B, or equal, shall be mounted on the left side of the truck on the front bumper plate. An electric brake control button for the siren shall be mounted on the truck dashboard. Controls for the siren shall be two foot-operated buttons, one for each the driver and the passenger. Two chromium-plated tone adjustable, trumpet-type air

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pressure shall be calculated as temperatures specified in 4.3.1.1. Net pump

horns shall be mounted on the cab roof, one on either side of the rotating warning light. The horns shall be controlled by a pull chain equally accessible to the driver and passenger seats. Air for the air horns shall be supplied by a line directly from the air tank. A buzzer shall be installed in the cab, and a waterproof pushbutton switch shall be mounted at the rear of the truck to provide a signaling device between the rear platform and the driver.

3.10.6 Bumper. A heavy-duty bumper shall be provided on the front of the truck chassis and shall be attached directly to the chassis frame. Provision shall be made at the front of the truck for storage of a 12-foot section of 6-inch soft suction hose. The storage bin shall be provided with drain holes.

3.10.7 Towing hooks, rings, or eyes. Towing hooks, rings or eyes, shall be provided to permit towing of the truck and towing of other apparatus by the truck. The contractor shall certify to the adequacy of towing devices.

3.10.8 Shock absorbers. Double-acting hydraulic shock absorbers shall be provided at both front and rear suspension which are matched to the suspension capacity.

3.10.9 Springs. Front and rear springs shall sustain the distributed gvw without evidence of overload or permanent set. The springs shall be designed and suspended for the superimposed loads so as to absorb shock and limit sway. Individual wheel loadings, with the truck loaded as specified in 3.9, shall be not greater than the rated capacities of the respective springs measured at the ground.

3.10.10 Truck drive assembly. The entire propulsion train between the engine and the drive wheels shall have a torque capacity at least equal to the maximum torque developed by the engine through the intervening speed and torque ratios.

3.10.10.1 Transmission. The transmission shall be of the full automatic type with not less than four forward and one reverse speed. The transmission shall be rated for heavy-duty fire service and designed to match the engine torque and speed to load demand. The transmission shall provide for driver selection of individual gears or range of gears required to meet performance requirements specified herein. The shift control shall be mounted on a pedestal to the right of the driver's seat and incorporate a positive indicator for each gear or range selected. The transmission shall be protected so that the operator cannot inadvertently shift into reverse. An electrically operated back-up warning system conforming to SAE J994 shall be installed to automatically operate when the transmission is put in the reverse mode. When the transmission is used to drive the fire pump, a locking device shall be installed to insure that the transmission and shift lever remains in the correct gear range.

3.10.10.2 Drive axles. The drive axle shall be of the full-floating type.

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3.10.11 Steering mechanism. Provisions shall be made for rapid adjustment of lost motion in the steering mechanism. The steering wheel shall be of ample size, shall not interfere with any other projecting equipment or operation of the vehicle, and shall be mounted on a rigid and sturdy column. Front wheels shall have a cramping angle of not less than 28 degrees right and left at the wheel on the inside of the turning circle. The truck shall be equipped with a power-assisted steering mechanism of the hydraulic type. The mechanism shall preserve, to the maximum practicable extent, the features of directional stability inherent in the manually operated steering mechanism, shall permit manual steering in the event that failure of the hydraulic system occurs, and shall be effective at all vehicle speeds.

3.10.12 Brakes. Brakes shall be standard for the truck furnished and as specified herein for the size vehicle furnished.

3.10.12.1 Service brakes. The truck shall be equipped with straight-air 4-wheel self-energizing-type service brakes. Service brakes shall control and hold the truck, loaded as specified in 3.9, on a 30 percent grade. Under all conditions of loading, service brakes shall stop the truck within 30 feet from a speed of 20 mph on dry hard level surfaces free from loose material. The brake mechanism shall be accessible for exterior adjustment. Brake lines shall be anchored to the chassis and protected from damage. Provisions shall be provided to prevent system freezing. All brakes shall conform to the latest DOT requirements.

3.10.12.2 Parking brake. The truck shall be provided with a parking brake. The parking brake shall be the self-energizing air-release type. The parking brake shall be controlled by a push-pull-type control on the cab dashboard. The air system when charged to not less than 100 psi shall not leak more than 25 percent in 24 hours.

3.10.12.3 Engine brake. The truck shall be equipped with an engine exhaust brake system designed for the diesel engine used. Braking shall be accomplished by motion transfer through a master-slave piston arrangement which opens the cylinder exhaust valves near the top of the normal compression stroke, releasing the compressed cylinder change to exhaust. An "on-off" dash switch and the accelerator switch shall control the engine brake. The brake shall be energized when pressure is removed from the accelerator pedal.

3.10.13 Wheel and tire assemblies. All components shall comply with Tire and Rim Association for 50 to 60 mph speed; tires at least 100 level quality; wheels inherently in balance. Tire and wheels shall be interchangeable front to rear. A spare tire and wheel shall be provided and shipped as an overpack, secured in the hose bed.

3.10.13.1 Single front wheels. Ten hole disc wheels with highway tread tires sized for the load imposed shall be provided.

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3.10.13.2 Dual rear wheels. Ten hole disc wheels with traction lug tires sized for the load imposed shall be provided.

3.10.13.3 Valves and caps. Valves to clear both wheels and brakedrum; valve caps and heat resistant T&RA G-4 cores shall be provided.

3.10.14 Angles of approach and departure. The angles of approach and departure shall be not less than specified by NFPA. The back step shall not be higher than 25.5 inches above ground with the truck loaded as specified in 3.9.

3.11 Cab. The cab shall have an upholstered seat and back. The engine compartment and the driver-passenger compartment shall be separated by an insulated firewall. The cab shall be closed construction. All wiring and tubing connected to roof mounted lights and horns shall be run behind a headliner or false roof. The cab shall be equipped with all tinted glass. A hot-water heater shall be provided. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. Heaters shall be complete with blower(s), and controls shall be mounted convenient to the driver. Floor mats and rubber boots shall be furnished for all lever openings through the floor. Instruments and controls shall be marked to identify their function and shall be installed to facilitate removal and servicing. Chassis equipment shall be complete with all accessories and shall include, but is not limited to, the following equipment:

(a) West coast type outside rear view mirrors mounted on each side of the cab. Each mirror shall be of the combination straight and convex type. Stick on type convex mirrors are not acceptable. The mirrors shall be provided with fold back type support arms. Mirrors shall be mounted so as to allow survey of both sides of the truck by the driver. Mirror vision shall not be affected by truck operation.
(b) Ammeter.
(c) Fuel gage.
(d) Oil pressure gage.
(e) Air pressure gage and low-pressure indicator light and buzzer.
(f) Transmission oil-temperature gage.
(g) Engine temperature gage.
(h) Hourmeter. The meter shall indicate by hands or counters not less than 9999.9 hours. The meter shall operate only when the engine is running.
(i) Speedometer with recording odometer.
(j) Tachometer. A tachometer shall be furnished in the dash of the truck, visible to the driver. A tachometer shall also be furnished on the pump panel. Both tachometers shall show engine speed over an operating range from 0 to 4000 rpm.
(k) Dual sun visors.
(l) Interior lighting, including jump seat area.
(m) Adjustable, weather-resistant, safety seat belt, with seat belt anchors. Seat belts shall have a noncorrosive metal-to-metal quick-release buckle for each seated occupant. Seat belt retractors shall be provided.

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3.1 Cross-lay hose beds. Two cross-lay hose beds shall be located over 3.1

(n) Jump seat areas shall be provided with a movable safety bar across the jump seat entranceway.
(o) Grab bars or hand rails shall be provided on the inside of the doors to aid personnel in entering the cab. Hand rails of the type specified in 3.13.4 shall be provided on the cab to the rear of each door to aid in entering the cab.

3.11.1 Custom cab. The front portion of the cab shall have an individual adjustable seat for the driver and a seat for the passenger. The front portion of the cab shall be partitioned off from the jump seat area by a sliding glass window. The rear portion of the cab shall house two jump seats, one on either side of the engine cover. The jump seats shall be not less than 22 inches wide. The jump seats shall face the rear of the truck. The top and sides of the cab shall extend equally to cover the jump seat area. The jump seat area shall be provided with an overhead dome light and control switch. The back rest of each jump seat shall have a padded upholstered quick-release type adjustable bracket to hold a 30 minute breathing apparatus. The flat seating area shall be not less than 16 inches from front to rear with the apparatus mounted.

3.11.2 Windshield wiper and washer. Windshield wipers shall be mounted to facilitate clear vision. Wiper assemblies shall include features to permit speed control and return of the wiper blade to a retracted position out of the line of vision when not in use. The windshield cleaning system shall include a device that will spray the path of the wiper blades with water or cleansing solution. The cleansing solution reservoir shall have a minimum capacity of 2 quarts. The windshield wiping/cleaning system shall conform to SAE J198.

3.12 Fire pump. The fire pump shall be midship mounted, driven by the truck engine. The pump transmission shall provide means for pump and road operation. The pump shall be of the centrifugal type. The pump shall have a certified UL standard 822 minimum rating of 1000 gpm at altitudes of 2000 feet. The pump drive, gearing, and housing shall be such that at rated capacity and pressure the temperature rise in the gear oil will not exceed the maximum temperature recommended by the pump manufacturer for continuous operation as specified herein. When recommended, oil cooling shall be provided (see 6.2). Draincocks, accessible at the pump operator's position and arranged so that the open and closed positions are clearly indicated, shall be provided to drain the pump and all water-carrying lines. All threaded connections shall conform to FED-STD-H28. The pump, when drafting through 6-inch suction hose, shall meet the pumping requirements of NFPA 1901.

3.12.1 Priming device. The priming device shall operate from the pump panel. The priming device shall be standard for the pump furnished and shall meet UL requirements.

3.12.2 Suction connections. The fire pump shall be provided with three 6-inch suction inlets, located one on each side of the truck and one located as a front intake. The inlets shall not be more than 4 feet above the ground, with a removable and accessible brass strainer provided inside each inlet. The front

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intake shall be fitted with a valve. The valve shall have a positive leakproof seal and shall be equipped with a hand adjustable control handle. Each side suction connection shall have a chromium-plated long-handled gasketed cap. The suction manifold shall also have a connection for intake from the booster tank. The tank suction line shall be sized to flow not less than 500 gpm while pumping. Suction inlets shall have machined ends so as to provide a seat for suction hose gaskets and shall be threaded with 6-inch NH threads. Two additional 2-1/2-inch gated gasketed female swivel suction inlets, threaded with 2-1/2-inch NH threads and equipped with chromium-plated rocker lug plug and chain, shall be provided on the street panel side of the truck. Sufficient clearances shall be provided to allow the use of spanner-type wrenches. The 2-1/2-inch suction connections shall be fitted with suitable sized, balanced, corrosion-resistant, full-flow-through-type valves and shall be operated and locked by means of an individual control handle as a single-handed operation.

3.12.3 Discharge connections. The truck shall be provided with the following discharge connections. Each discharge shall be provided with a suitably sized, balanced, corrosion-resistant, full-flow-through-type valve. Each valve shall be operated and locked by means of an individual control handle as a single-handed operation from the pump panel.

(a) Two 2-1/2-inch NH outlets on each side of the truck. Each outlet shall be fitted with a 2-1/2-inch 30 or 45 degree elbow and 2-1/2 x 1-1/2-inch NH adaptor with 1-1/2-inch cap and chain.
(b) One NPT connection piped to the booster reel.
(c) One discharge for connection with the deck pipe sized to flow not less than 750 GPM.
(d) One 1-1/2-inch discharge from the pump to the tank.
(e) Two NPT discharges from the pump to the two 1-1/2-inch NH hose bin over the pump sized to flow not less than 500 GPM with an internal pressure drop of not more than 35 psi. Each outlet shall be fitted with a 1-1/2-inch fitting..
(f) Two NPT discharges from the pump to the rear hose bins and sized to flow not less than 500 GPM with a internal pressure drop of not more than 35 psi. Each outlet shall be fitted with a 2-1/2-inch NH fitting and a 2-1/2 to 1-1/2-inch NH adaptor.
(g) One discharge guage for the water tower (when a water tower is specified).

Each discharge specified in (a) and (c) above shall be provided with a drain valve of at least 3/4-inch inside diameter for draining hose lines with the discharge gates closed. Drain valve controls shall be located adjacent to the discharge control. Drain lines shall be concealed behind the pump compartment panels and shall discharge to a point not above the bottom of the chassis frame. Controls for the 2-1/2-inch discharges on the right side of the truck shall be of the piggy back remote control type operable from the right side of the truck and from the pump control panel. Each discharge outlet shall have machined ends so as to provide a seat for the hose gasket and shall be provided with a rocker lug fitting. The discharge valves shall assure ease of operation under all operating conditions. Clearances shall be provided to allow the use of

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spanner-type wrenches. The two 1-1/2-inch hose bins over the pump shall be fitted with a swivel to allow playout of the hose from either side of the truck.

3.12.4 Relief valve. One automatic relief valve shall be provided for controlling the pressure from all outlets. The location of the relief valve control shall be such that adjustment can be made by the operator when standing in normal pump operator's position, without the use of tools, and in full view of the discharge pressure gage. The relief valve shall be of such design that, when set 10 psi higher than the pump working pressure, there will not be an increase in pump pressure greater than 30 psi when discharge lines are shut slowly. The relief valve shall perform satisfactorily when pumping rated capacity at any pump working pressure is over 75 psi. An indicator light shall be furnished on the control panel.

3.12.5 Controls . Provisions shall be made for quick and easy operation of the fire pump. All levers or other devices used in connection with pumping operations shall be provided with positive catches or other means to prevent them from being jarred or moved out of position. All controls for changing from road transmission to pump transmission shall be marked for road and pumping positions. All pumping controls and instruments necessary to operate the fire pump under all conditions of service shall be arranged on a stainless-steel pump panel located on the left side of the truck, and located so that all hose-line operations and pump-work adjustments necessary during service may be effected without maneuvering by the operator. The throttle control shall be of the screw, center-push-button-release type and shall control the entire range of engine speed smoothly. The throttle control shall hold the control lever in any designed position and shall not be affected by vibration or by pull of the accelerator spring. Controls and gages shall be as symmetrically grouped as practicable. Plates or collars shall be installed around all discharge outlets and controls on the control panels when required. Each discharge control shall be systematically numbered. The assigned number shall be mechanically adhered to the pump panel near the controls. Mechanical means shall be provided to engage and disengage the pump under emergency conditions. Controls shall be marked and shall be accessable to the pump operator without getting under the truck or using tools. When a water tower is required a pull-out treadplate step under the pump panel on which the pump operator can stand is required. The step shall support not less than 300 pounds on the outside edge. The step shall be secured in a close position when not used.

3.12.6 Gages and instruments. All gages shall have provisions for calibration. The master pressure and compound gage shall have a dial face not less than 3-1/2 inch diameter. The suction gage shall have a dial pressure range of 30 inches of Hg to a pressure of not less than 300 psi. The master discharge gages shall have a dial pressure range of 30 to 600 psi. Each fire hose discharge outlet specified in 3.12.3, except (b) and (d), shall have a separate pressure gage of not less than 2-1/2 inch diameter. Each gage shall be provided with a device to eliminate pressure fluctuation, glycerin filled gauges are acceptable. The master discharge gage shall be connected to the pump manifold and the remaining gages connected to the hoseline side of each of the

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discharge valves. Gages shall be systematically arranged and numbered to correspond with the controls. Numbers shall be affixed as specified in 3.12.5. Throttling valves and drain controls shall extend to the exterior of the pump control panel. An indicator or gage shall be provided to show the water level in the water tank specified in 3.14. Valve bodies and connecting piping shall be concealed behind the pump control panel. Two plugged 1/4-inch NPT connections for test gages, one connected to the pressure side and one to the suction side of the pump, shall be mounted on the pump panel. Each connection shall be marked "Pressure" or "Suction". The tachometer on the pump panel shall indicate the engine speed when driving the pump. Provisions shall be made at the pump panel for checking the tachometer with a hand revolution counter without disconnecting the tachometer. All gages shall be moisture tight to prevent the entrance of water or of fogging due to temperature change. Engine operation gages for oil pressure, water temperature and ammeter shall also be installed in the pump panel.

3.13 Body. The truck shall be equipped with a body which shall incorporate a pump housing, operator's panel, hose bed, equipment cabinets, running boards, rear step, and water tank. Step plates shall be furnished on each side at the rear of the truck to aid personnel to move from the rear platform to the hose bed. The body shall mount the fire pump and stow the fire fighting equipment specified herein. The body shall be reinforced at all points where hose and ladder clamps and hangers are attached.

3.13.1 Pump housing and stowage bin. A top stowage bin shall be provided over the pump. This bin shall have a removable metal panel bottom to enable the pump to be removed with the suction and discharge connection removed. Drain holes shall be provided to eliminate water retention. The stowage bin bottom shall support 200 pounds on any 1 square foot area without deformation. The bottom shall be fitted with a slated floor as specified in 3.13.2. The bin shall run the full width of the truck and shall be divided into two equal bins for stowage of 1-3/4-inch preconnected hose lines. Each bin shall hold not less than 200 feet of hose. The sides and bottom opening of each hose bin shall be provided with hose rollers to aid in playing out hose. Provisions shall be made for mounting the lights specified in 3.25(j). Electrical outlets specified in 3.10.5.2.1 shall be provided for connecting each light.

3.13.2 Fire hose compartments. Compartment for stowing a minimum of 1200 feet of 3-inch double-jacketed cotton rubber-lined fire hose shall be provided. The 3-inch hose compartment located between the wheel housing shall be divided lengthwise with a metal partition to form two equal hose beds. Two leader hose compartments shall be provided, one on either side of the 3-inch hose bed. The leader bed shall hold not less than 300 feet of 2-1/2-inch hose. Body panels and partitions shall be of the same height. The compartment body and partitions shall be reinforced so as to offer no obstruction to playing out or loading of hose. The compartment floor shall be closed with sheet metal to provide protection to the hose from road dirt and tire splash. Drain holes shall be provided to preclude water retention. The metal floor shall be reinforced by a crossmember which shall support a slatted floor of seasoned select-grade

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hardwood or aluminum. Slats shall be not less than 3-1/2-inches in width. The slats shall have a finished thickness of not less than 3/4-inch, shall run the full length of the compartment, and shall be spaced so as to provide ventilation for hose. The spacing shall be not less than 5/8-inch nor more than 3/4-inch. The slats shall be fastened to the crossmembers by countersunk screws or bolts. The slatted floor shall be removable. The interior of the compartment shall be free from all projecting nuts, sharp angles, or brackets. Exposed bolt or rivet heads shall be of the oval type. The tank vent or overflow shall not extend into the hose bed. Hose beds shall be provided with close fitted covers made of Herculite or equal material. The covers shall be fitted with matching fasteners for attaching to the truck body. The cover shall be designed to allow playout of the hose without damage to the cover or disconnection of any fasteners.

3.13.3 Equipment cabinets. Equipment cabinets shall be provided on each side of the truck. The right or curb side of the truck shall have three compartments, one in front of the rear wheel well, one to the rear of the wheel well, and a through compartment in front of the pump compartment (space). The through compartment shall be not less than 8 inches wide and shall be common to both sides of the truck. The compartment shall be not less than 24 inches high with a removable shelf. The left or street side of the truck shall have the same compartment arrangment with three additional compartments mounted above the bottom compartments. The top compartments shall run the full length of the bottom compartments and shall have a height equal to the distance from the top of the bottom cabinet to the top of the hose bed. The top cabinets shall be deep enough to store complete 30 minute air pack mounted in vertical brackets but shall not be less than 12 inches deep. Packs and brackets shall not be furnished. All cabinets shall be recessed not less than 3/4-inch so that they will not be flush with the running boards or wheel skirtings. All bottom cabinets shall be not less than 25 inches in height; cabinets depth shall be the same as the wheel housings, but not less than 20 inches. All doors shall be of box type construction and shall be gasketed. Cabinets shall be provided with vent openings to eliminate sweating inside compartment spaces. All cabinets shall have slatted flooring as specified in 3.13.2. Each cabinet door shall be provided with a recessed "D" ring-type handle and flush bolt-type lock(s). Doors shall operate on corrosion resistant piano type hinges. Hinges shall run the full length of the door. All cabinets shall be illuminated, and the lighting shall be automatically controlled through door switches to light when the doors are opened. Cabinet tops shall run continuously on each side of the truck and shall be of antiskid surface similar to the running board and rear step surface.

3.13.4 Handrails and hose rollers. The rear step shall be provided with two vertical grabrails for mounting the step, and a handrail not more than 66 inches above the step. The handrail shall be located not more than 6 inches forward of the rear end of the hose bed, or 6 inches forward of the clear area of the rear step. A horizontal, ribbed-rubber covered handrail shall be located just below the hose bed. Hose pickup lights shall not be mounted or attached to the handrail. Both the grabrails and handrails shall be not less than 1-1/4-inch

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heavy tubing. All equipment attached to the body within reach of a running board shall be attached so as to serve as a personnel handrail. Two additional handrails of 1-1/4-inch heavy tubing shall be provided on top of the hose compartment at the right and left sides, and shall be the same length as the compartment sides. Vertical handrails of 1-1/4-inch heavy tubing shall be provided on both sides of the jump seat entranceway and cab doors. Handrails shall be mounted on brackets and shall not be free to turn in the brackets. Hose guide rollers shall be provided for the booster hose reels and for both sides and bottom of the two 1-1/2-inch hose beds over the pump compartments. Hose rollers shall be chromium-plated in accordance with QQ-C-320 Class J, Type I on CRES stainless steel that has been passivated in accordance with QQ-P-35. Hose rollers shall not serve as handrails.

3.13.5 Fasteners. Fasteners used in removable panels, inspection plates and compartment closures shall be of the quick-disconnect type. Sheet-metal screws are not acceptable.

3.14 Water tank and piping. A removable water tank of not less than 500 useable gallon capacity shall be mounted on the truck frame to the rear of the pump. The tank shall not extend beyond the hose bed sides. The tank shall be of welded construction. The entire tank top shall be removable. The top retaining capscrews or fasteners shall not come in contact with the water. Tank construction and outlet shall allow the water to be pumped out when the truck is standing on a 20 percent grade in all planes. The tank shall be baffled in accordance with NFPA 1901 to prevent excessive surging of the water. The tank shall be equipped with a fill opening having an area of at least 64 square inches to permit easy filling of the tank. The cover shall be gasketed and retained in place to prevent spillage during operation. The tank fill openings shall be provided with a removable screen to prevent entrance of foreign matter. The screen shall have screen openings not larger than 1/4 square inches. Water shall not surge from a filled or partially filled tank due to acceleration, deceleration, or movement of the truck. The tank shall have a nonsplash combination overflow and vent pipe extended to the rear of the truck and directed inboard to the ground so that water from the overflow drain will not affect traction of rear wheels. The tank shall be constructed of stainless steel type S30403 per ASTM A240 and shall be freed of mill and weld scale either by a pickling treatment or sandblasting. All piping from the tank and pump unless made of corrosion-resisting metal, that has been passivated per QQ-P-35 shall be galvanized to a G90 coating designation. The pipeline between the tank and the pump may incorporate a flexible reinforced noncollapsible connection(s) to take up vibration, and shall be provided with a quick-opening shutoff valve with control extended to the operating side of the truck and a drainvalve located at the lowest point with control extended to the operating side of the truck. The tank-to-pump line shall include a check valve to prevent the flow of water from the pump to the tank. The tank suction pipeline valves and fittings shall withstand not less than 120 psi working pressure. The discharge pipeline, valve, and fittings between the pump and booster reel shall withstand not less than 350 psi working pressure. A complete piping diagram shall be provided with the manuals furnished with each truck. The tank shall show no evidence of leaking after being subjected to the tests specified in 4.5.2.

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3.15 Booster hose reel. The booster hose reel shall be provided with 150 feet of 1-inch rubber chemical hose jointed with smooth full-flow-type couplings. The reel shall have a friction brake to prevent overtravel and shall be an electric-rewind type. The reel shall also include provision for handcrank rewind. The reel shall be mounted in a compartment in front of the rear platform between the truck frame. The hose reel compartments shall be provided with drains and a roll up door (see 3.21.7).

3.16 Fire fighting equipment mounting. The truck shall be provided with the necessary holders, brackets, mounting pads, and tool chests to attach the fire fighting equipment specified in 3.25. The attachments shall be of the type that remain securely in place under all running conditions and yet permit quick removal without interference with other equipment. No item of fire fighting equipment specified herein shall be stowed aboard the truck without a heavy-duty method of attachment. Such externally mounted mounting attachments shall, whenever possible, be chromium plated. All attachments shall be provided with spacers for mounting away from running boards or body to allow for drainage. The two 10-foot lengths of 6-inch noncollapsible suction hose (see 3.25), shall be mounted and stowed in metal tubes between the hose bed (see 3.13.2) and the main water tank (see 3.14). The tubes shall be made of corrision-resisting stainless steel that has been passivated in accordance with QQ-P-35, and shall be shaped to permit ventilation between hose and tube over the entire length without chafing the hose. The male end of each hose shall face the front of the truck. Access to the hose shall be from the rear of the truck. The tubes shall be designed to eliminate water retension and an access door to keep out rain and road dirt. Locks or spring-loaded devices shall be used to hold the hose under all road conditions. Hose tubes shall be removable, when necessary, in order to remove the water tank top. All ladders shall be mounted on the right side of the truck above the cabinets and wheel-housing at a height which will not interfere with operation of pump valves. The 14-foot roof-type ladder and the 36-foot extension ladder (see 3.25), shall be mounted together. Ladders shall be mounted by mechanical all- metal quick-release adjustable double-control fastenings that will prevent chafing and rattling. The 10-foot laterally folding attic-type ladder (see 3.25), shall be mounted on top of the cabinet surface below the other ladders. The ladder shall be held in place by an all-metal fastener. All ladders shall conform to NFPA No. 1931. Ladders mounting height shall be held to the lowest practicable height above ground.

3.17 Mechanical foam system. The truck shall be equipped with foam-producing equipment. The foam proportioner shall be of the around-the-pump type. A metering valve shall be provided in the foam liquid line for controlling the amount of foam liquid. The foam liquid metering valve shall be calibrated and shall be provided with a dial and pointer to regulate the proportioning or discharge percentage of foam liquid to the total solution flow. A check valve shall be installed to prevent flow of water back into the foam tank. Provisions shall be made to flush out the foam system after use. A valve shall be provided

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in the line from the foam tank to the metering valve. The valve shall be installed as close to the foam tank as possible. The system shall deliver 3 to 8 percent foam solution through the normal operating pressure range of the fire pump and at discharge capacities of up to 1000 gpm.

3.17.1 Controls. All controls for operation of the foam system shall be located on the fire pump instrument panel and shall be accessible to the operator. An instruction plate showing the proper setting of the metering valve shall be located adjacent to the foam controls.

3.17.2 Foam tank. The foam liquid tank shall be located in the area of the pump housing. The tank shall have a capacity of not less than 60 useable gallons. The tank shall be constructed of stainless steel type S30403 in accordance with ASTM A240 and shall be free of mill scale either by a pickling treatment or sandblasting. The tank shall be removable from the body. Baffles shall be installed to prevent excessive surging and foaming of the liquid. A drain with discharge at a point below the bottom of the truck frame, and readily accessible from one side of the truck, shall be provided with a valve control located at the fire pump control panel. The drainvalve shall be located immediately adjacent to the tank so as to minimize the amount of piping between the tank and valve. The foam system shall be installed as suggested by the foam equipment manufacturer. Provisions shall be made for filling from a 5-gallon container. The filler neck shall extend far enough into the tank to reduce foaming overflow while filling. The filler opening shall be large enough to allow hand cleaning of the inside of the tank. A splash-proof vent shall be provided in the foam tank and shall be piped to the outside of the truck. Removable screens shall be provided, one at the filler neck and one on the outlet of the foam tank, that will prevent entry of foreign matter into the proportioner or pump. The filler shall be provided with a gasketed-latching-type cover. The tank shall be fitted with an electronic type liquid level indicator or gage. The indicator or gage shall be mounted in the pump panel as specified in 3.12.6.

3.17.3 Piping. All piping and valves coming in contact with foam liquid shall be of a corrosion-resistant material for use with foam liquid conforming to MIL-F-24385. Valves shall be installed between the foam liquid tank and metering valve and between the pump discharge and eductor. A check valve shall be installed between the eductor and metering valve, and a horizontal swing check valve shall be installed in the water-tank suction line to prevent entry of foam solution into the water tank.

3.18 Communication equipment. When specified (see 6.2), a radio communication system, or a space approximately 1-1/2 foot long x 1-1/2 foot wide x 1 foot high for installation of communication equipment, shall be provided.

3.19 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission and susceptability characteristics of the truck shall be certified to SAE J551.

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3.20 Lubrication. Unless otherwise specified (see 6.2), the truck shall be lubricated prior to delivery with lubricants designated for use in the -25° F to 125° F temperature range. The truck shall operate as specified herein when serviced with military oils designated for use in the temperature range specified herein (see 6.7). A tag shall be attached to the transmission control level to indicate the temperature range, and the lubricant grade used.

3.21 Water tower. When specified (see 6.2), the truck shall be furnished with a 50 foot water tower with ladder attached.

3.21.1 Nozzle. A remote control nozzle shall be provided at the end of the tower. Controls for the nozzle shall be at the tip of the tower and the base control panel to actuate elevation, sweep and stream pattern of the nozzle. The nozzle shall automatically adjust to flow rates from 300 to 1000 gpm to provide a good stream of water. Nozzle shall be capable of being elevated in a vertical plane with travel of 250 degrees and have lateral sweep of 90 degrees (45 degrees either size of center) at any angle of elevation. The lateral sweep shall be controlled by a hydraulic cylinder and elevation by a hydraulic monitor. Function of elevation and sweep shall be independent of each other and controlled by a single switch or lever so that the range of sweep, in a plane, is the same regardless of the position of the boom or elevation of the nozzle. The underside of the outer end of the boom shall be provided with a recess whereby the nozzle can be retracted into the recess when not in use. Nozzle shall be protected from damage when retracted without the use of external guard rails on the underside of the boom.

3.21.2 Water boom and ladder combination. The boom shall be constructed of boxbeam aluminum, telescoping section. The water tower/ladder shall have an operating height of not less than 49 feet. The tower shall have a horizontal reach at 0° over the side of the truck of not less than 40 feet. All measurements shall be taken in accordance with NFPA 1901. The design of the water tower shall allow boom extension, retraction, raising and lowering while flowing water at 1000 GPM.

3.21.2.1 Performance of water boom and ladder combination. The boom shall be so constructed and powered that it will be capable of performing the following functions.

(a) With boom fully extended, tip unsupported, flowing water up to 1000 GPM without men or load on the ladder, support and resist the reaction of the nozzle in any normal position of operation with the boom at any angle from 10 degrees below horizontal up to 90 degrees above horizontal.
(b) With boom fully extended and raised to any angle between 45 degrees and 75 degrees tip unsupported and flowing water up to 1000 GPM, support and resist the reaction of the nozzle in any position of operation plus a 300 pound load at ladder tip.

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(c) Used as a ladder only (no water flowing) with boom fully extended, tip unsupported and raised to any angle from 0 degrees to 45 degrees, support a tip load of 400 pounds.
(d) Used as a ladder only (no water flowing) with boom fully extended, tip unsupported and raised to any angle between 45 degrees and 75 degrees, support a tip load of 800 pounds.
(e) Used as a ladder only (no water flowing) with boom fully extended, tip supported, cary and support a load of 1600 pounds evenly distributed along the length of the ladder.

3.21.3 Turntable. A turntable not less than 36 inches in diameter, able to rotate in continuous 360° turns in either direction shall be provided. The turn table shall be controlled from the rear of the truck and shall be operated by a hydraulic motor.

3.21.4 Water system. Water for the tower shall be supplied by the truck pump or by two 2-1/2-inch inlet/outlet valves at the rear of the truck. A 4-inch valve shall control water to the water tower. The tower shall be able to discharge 1000 gpm at any position with the truck setting on level, hard surface with the outrigger in the operating position.

3.21.5 Outriggers. The tower shall be furnished with hydraulically operated outriggers. The outriggers shall be self-locking and shall be arranged to stabilize the truck when the tower is raised and flowing water. Controls shall be provided to render the outriggers inoperative when the boom is in raised position.

3.21.6 Control station. A control station shall be provided for the water tower-ladder at the rear of the truck. The station shall provide for:

(a) One-hand control of the tower.
(b) Control of the nozzle.
(c) Control for outriggers.
(d) Master electric switch.
(e) Emergency electric hydraulic system controls.
(f) Water pressure guage.
(g) Throttle control switch.
(h) Water valve control.
(i) Communication system to tip of ladder.
(j) Complete remote control system for nozzle.

3.21.7 Booster hose reel. The booster hose reel, compartment and associated piping and controls specified in 3.15 shall not be furnished when a water tower is specified. The two noncollapsible suction hoses and storage tubes specified in 3.16 shall not be furnished when a water tower is specified. The deck pipe, piping and controls specified in 3.25(f) shall not be furnished when a water tower is specified (see 6.2).

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3.22 Identification marking. The truck shall be identified in accordance with MIL-STD-130. The marking shall be applied to the truck on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 Material. Each plate shall be attached by screws, bolts, or rivets to the inside of the door on the driver's side of the cab. The following data shall appear in the ownership and nomenclature spaces provided on the identification plate in their respective order:

OWNERSHIP - U.S.
NOMENCLATURE - TRUCK, FIRE FIGHTING, 1000 GPM 4 x 2

3.23 Instruction plates. All pumping controls, valves, instruments, and truck operating controls shall be provided with instruction plates designating function and operation. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material, size as applicable. Each plate shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.24 Painting. Unless otherwise specified (see 6.2), the truck cab, chassis body and other equipment, as applicable, shall be cleaned, treated, primed, and painted two coats gloss lime-yellow, color No. 13670 of FED-STD-595 in accordance with the manufacturer's standard commercial practice, but not less than equal to MIL-T-704. The ladder, turnable and all tread plates shall be painted aluminum color. The vehicle shall be rust-proofed in accordance with FED-STD-297.

3.25 Contractor-furnished components. The following equipment shall be furnished, installed, and stowed on each truck:

(a) Three 50-foot lengths of 1-inch inside diameter smooth-bore, rubber-lined hose shall be installed on the hose reel specified in 3.15. Each length shall be fitted with reusable couplings. Hose shall meet NFPA requirements and have not less than a 600 psi working pressure rating. A 1-inch NH combination fog and straight-stream nozzle shall be installed on the hose and mounted in a quick-release device. The nozzle shall be the ball shutoff type with adjustable pattern and gallonage control (see 3.21.7).
(b) Two 10-foot lengths of 6-inch hard suction hose, commercial grade, NFPA approved, with 6-inch NH long-handled couplers on female ends and 6-inch NH rocker lug couplers on the male ends. The hose shall be mounted in tubes specified in 3.16 (see 3.21.7).
(c) One 12-foot length of 6-inch soft suction hose, commercial grade, NFPA approved, with a 4-1/2-inch long handled swivel female gasketed coupling on one end and a 6-inch long handled swivel female gasketed coupling on the other end. The hose shall be mounted in a storage bin on the front of the truck.

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(d) One strainer with a 6-inch NH internal coupling shall be attached to a mounting pad on the step plate at the pump panel. Furnished only when the hand suction hose is furnished.
(e) Ladders.

(1) One 14-foot roof-type ladder.
(2) One 36-foot three-section extension ladder.
(3) One 10-foot laterally folding attic-type ladder.
All ladders shall be in accordance with NFPA No. 1931.

(f) One deck pipe, Akron style 506 or equal, for discharge gallonage not less than 1000 gpm. Handle control operation for change to any horizontal or vertical position. Horizontal and vertical locks. Furnished with a 1000 gpm variable stream fog nozzle Akron style 1750 or equal (see 3.21.7).
(g) Three straight plain point pike poles, with hook and fiberglass handle: One 6 feet, one 10 feet, mounted in tubes beside the attic type ladder, and one 32 inches with "D" - type handle grip mounted to the inside of the through compartment curb side door.
(h) Two axes: One pick-head fire axe with a fiberglass handle; and one flat-head Dayton pattern fire axe with a fiberglass handle, mounted one on either side of the truck on the outside cab jump seat area.
(i) One reducer coupling hose, double internal, long handled, double swivel, female, gasketed couplings, the size shall be 4-1/2-inch NH to 6-inch NH mounted to the inside of the streetside through compartment door.
(j) Two portable floodlight 300 watt, with three wire ground connector. Circle-D portable light No. 150 and model P-23 holddown clamps or equal. The lights shall be mounted one on either side, near the side of the truck over the pump compartment.

3.26 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.27 Workmanship.

3.27.1 Castings and forgings. All parts, components, and assemblies of the truck which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

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3.27.2 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or splatter.

3.27.3 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.27.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the truck to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16-inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the truck shall be at the option of the contractor.

3.27.5 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

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3.27.6 Bolted connections. Boltholes shall be accurately formed and shall have burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.27.7 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.6).
(d) Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article truck shall be examined as specified in 4.5.1. The examination shall be conducted after all of the tests specified in 4.3.2 have been completed. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article truck shall be tested as specified in 4.5.2, except the pump test shall be conducted and certified by UL before government inspection and testing. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. Unless otherwise specified (see 6.2), one or more trucks will be selected at random by the Government from trucks being produced by production tooling and will be examined as specified in 4.5.1 and

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tested as specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production truck shall not exclude the remaining trucks from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production truck(s) and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all trucks accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.4.2 Tests. Each truck shall be tested as specified in 4.5.2.1, 4.5.2.1.2, 4.5.2.1.3, and 4.5.2.2. Failure of any test shall be cause for rejection. The test in 4.5.2.1.3 may be performed using a water dye in lieu of foam liquid.

4.5 Inspection procedure. Prior to examination or tests, the truck shall be lubricated with military oils and greases for use at the ambient temperature at the inspection site.

4.5.1 Examination. The truck shall be examined as specified for both first article and initial production inspection for the characteristics listed in table I.

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TABLE I. Examination schedule (requirement).

Number	Characteristic	Requirement Paragraph
1	2	3
Major		
101.	Truck not as specified.	3.1
102.	Material not as specified.	3.4
103.	Safety not as specified.	3.6
104.	Human factors not as specified.	3.7
105.	Truck chassis not as specified.	3.10
106.	The truck manufacturer's gross vehicle weight (gvw) rating for the truck chassis not as specifed.	3.10.1
107.	Wheel base not as specified.	3.10.2
108.	Length and width of the truck and equipment not as specified.	3.10.2.1
109.	The chassis frame not as specified.	3.10.3
110.	Engine not as specified.	3.10.4
111.	Engine fuel system not as specified.	3.10.4.1
112.	Lubrication of the engine not as specified.	3.10.4.3
113.	Exhaust system not as specified.	3.10.4.4
114.	Brake horsepower certification not furnished by contractor as specified.	3.10.4.5
115.	Electrical equipment not as specified.	3.10.5
116.	Ignition system not as specified.	3.10.5.1
117.	Battery units and electrical circuits not as specified.	3.10.5.2
118.	Lighting system not as specified.	3.10.5.3
119.	Additional lighting not as specified.	3.10.5.3.1
120.	Warning light not as specified.	3.10.5.3.2
121.	Light switches not as specified.	3.10.5.3.3
122.	Siren and horn not as specified.	3.10.5.4
123.	The specified type of bumper not furnished as specified.	3.10.6
124.	Towing hooks, rings, or eyes not as specified.	3.10.7
125.	Shock absorbers not as specified.	3.10.8
126.	Front and rear springs not as specified.	3.10.9
127.	Truck drive assembly not as specified.	3.10.10
128.	Transmission not as specified.	3.10.10.1
129.	Drive axle not as specified.	3.10.10.2
130.	Steering mechanism not as specified.	3.10.11
131.	Brakes not as specified.	3.10.12
132.	Service brakes not as specified.	3.10.12.1

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TABLE I. Examination schedule (requirement). (Continued)

Number	Characteristic	Requirement Paragraph
1	2	3
133.	Parking brake not as specified.	3.10.12.2
134.	Wheel and tire assembly not as specified.	3.10.13
135.	Single front wheels not as specified.	3.10.13.1
136.	Dual rear wheels not as specified.	3.10.13.2
137.	Valves and caps not as specified.	3.10.13.3
138.	Angles of approach and departure not as specified.	3.10.14
139.	Cab not as specified.	3.11
140.	Custom cab not as specified.	3.11.1
141.	Windshield wiper and washer not furnished as specified.	3.11.2
142.	Fire pump not UL rated as specified.	3.12
143.	Priming pump not furnished as specified.	3.12.1
144.	Suction connections not furnished as specified.	3.12.2
145.	Discharge connections not furnished as specified.	3.12.3
146.	Relief valve not provided as specified.	3.12.4
147.	Controls not as specified.	3.12.5
148.	Gages and instruments not as specified.	3.12.6
149.	Body of truck not as specified.	3.13
150.	Pump housing and stowage bin not as specified.	3.13.1
151.	Fire hose compartments not as specified.	3.13.2
152.	Equipment cabinets not as specified.	3.13.3
153.	Handrails and hose rollers not as specified.	3.13.4
154.	Fasteners not as specified.	3.13.5
155.	Water tank and piping not as specified.	3.14
156.	A booster hose reel not furnished as specified.	3.15
157.	Mounting for fire fighting equipment not provided as specified.	3.16
158.	Mechanical foam system not furnished as specified.	3.17
159.	Controls for operation of the foam system not located as specified.	3.17.1
160.	Foam tank not located as specified.	3.17.2
161.	Piping and valves coming in contact with foam liquid not of corrosive resistance material as specified.	3.17.3
162.	Communication equipment not furnished as specified.	3.18
163.	The truck not lubricated before delivery as specified.	3.20

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TABLE I. Examination schedule (requirement). (Continued)

Number	Characteristic	Requirement Paragraph
1	2	3
164.	Water tower not as specified.	3.21
165.	Instruction plates not made and furnished as specified.	3.23
166.	Other specified components not furnished by contractor.	3.25
167.	Interchangeability of parts having the same part number not as specified.	3.26
168.	Casting and forgings not as specified.	3.27.1
169.	Aluminum welding not as specified.	3.27.2
170.	Metal fabrication not as specified.	3.27.3
171.	Welded surface not prepared for welding as specified.	3.27.4
172.	Welders not certified for welding as specified.	3.27.5
173.	Bolted connections not formed and burrs removed as specified.	3.27.6
174.	Riveted connections not as specified.	3.27.7
Minor		
201.	Identification marking not as specified.	3.22
202.	Painting not as specified.	3.24

4.5.2 Tests.

4.5.2.1 Pump. The pump shall be tested and certified by Underwriter Laboratories to UL822 prior to any tests specified in 4.3. All gages shall be checked for accuracy of 10 percent of total gage range.

4.5.2.1.1 Hydrostatic. The entire pump, including the suction chamber and all piping, shall be tested to a hydrostatic pressure of 350 psi for 10 minutes. Nonconformance to 3.14 shall constitute failure of this test. This test shall be conducted after all tests specified in 4.5.2.2 have been completed.

4.5.2.1.2 Water tower. The water tower and ladder combination shall be tested for conformance to 3.21 through 3.21.6. Nonconformance to any condition shall be cause for rejection.

4.5.2.1.3 Foam system. The foam system shall be tested using foam liquid conforming to MIL-F-24385. The metering valve shall be set to deliver 3 percent of foam solution and then to deliver 6 percent of foam solution

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for not less than 1 minute at each setting while discharging 200 gallons of water per minute. Failure to use at least 6 gallons of foam liquid per minute at 3 percent and 12 gallons of foam liquid at 6 percent shall constitute failure of this test.

4.5.2.1.4 Test for ambient temperature and altitude conditions. When specified (see 6.2), the contractor shall perform a pumping test of the first article under actual or simulated conditions of the specified altitude above sea level and at an ambient temperature of 125° F. When pumping at heights above 2000 feet altitude is required, certification is required from the contractor that the vehicle will perform as specified in 3.10.4.2. The pumping test shall consist of the following:

(a) Test the suction and priming.
(b) The pumping test shall be for 3 hours consisting of 2 hours at rated capacity at 150 psi discharge gage pressure, 1/2 hour at 70 percent of rated capacity at 200 psi discharge gage pressure, and 1/2 hour at 50 percent of rated capacity at 250 psi discharge gage pressure. The test shall be conducted with 20 feet of 6 inch suction hose and a suction lift of not less than 10 feet.

Nonconformance to 3.12 shall constitute failure of this test.

4.5.2.2 Road tests. The loaded truck, including filled water tank, filled foam tank, filled gasoline tank, lubricants, a 1500 pound evenly distributed simulated load, and a crew of five men, weighing not less than 175 pounds each or equivalent dead weight, shall be tested as follows:

(a) 100 miles over paved roads at maximum traffic speeds up to 55 mph.
(b) 50 miles over graded dirt roads at speeds up to 30 mph.
(c) Service brakes shall be applied during runs (a) and (b) above.
(d) All lights, siren, horn, speedometer, hourmeter, controls, and instruments pertaining to the truck engine shall be tested.
(e) The airbrake system shall be tested for conformance to 3.10.12.2.

Nonconformance shall constitute failure of these tests.

4.5.2.3 Electromagnetic interference. The truck shall be tested as specified in SAE J551. Vendor certification may be used.

4.5.2.4 Sound level test.

4.5.2.4.1 Extension sound level. The contractor shall perform sound level tests in accordance with SAE J366. Certified prior test data is acceptable, if available. Nonconformance with 3.7.1 shall constitute failure of this test.

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4.5.2.4.2 Personnel noise level. The truck shall be tested in accordance with MIL-STD-1474. Typical duty cycle tests shall not be used. Failure to meet the requirements of 3.7.1 shall constitute failure of this test.

4.6 Quality conformance inspection.

4.6.1 Examination. Each truck shall be examined for defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.6.2 Tests. Each truck shall be tested as specified in 4.5.2.1, 4.5.2.1.2, 4.5.2.1.3, 4.5.2.1.4, 4.5.2.2, 4.5.2.4.1 and 4.5.2.4.2. The road test specified in 4.5.2.2 shall be reduced by one half the required mileage. Failure of any test shall be cause for rejection.

4.7 Inspection of packaging. The preservation, packing and marking of the truck and equipment shall be examined to determine compliance with the Quality Assurance Provisions of MIL-STD-281 and MIL-E-10062, as applicable.

5. PACKAGING

5.1 Preservation. The degree of preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each complete truck, with equipment shall be preserved in accordance with MIL-STD-281, Level A, (mobile), with the following additions and exceptions.

5.1.1.1 Engine. Engine and engine accessories shall be preserved in accordance with MIL-E-10062, Type I, Method I, Level A.

5.1.1.2 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure as specified in MIL-STD-281, except the antifreeze shall conform to MIL-A-46153.

5.1.1.3 Equipment. Equipment shall be secured as provided for, except exposed equipment subject to pilferage, damage or loss shall be prepared as "On vehicle equipment" in accordance with MIL-STD-281.

5.1.2 Level B. Each complete truck shall be preserved in accordance with MIL-STD-281, Level B (mobile), and with the additions and exceptions as specified for Level A.

5.1.3 Level C. Each complete truck and all equipment issued with the truck shall be suitably preserved to afford protection against deterioration and damage from the contractor to the initial destination. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are

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fragile by nature, shall be protected by wrappings, pack compartmentization, or cartonizing of the individual item. Exposed equipment subject to easy pilferage or loss shall be placed in secured compartments, truck cab or in wooden boxes. Wooden boxes, when used, shall be secured to the truck with metal strapping and in a manner to prevent damage to the painted surfaces of the truck. The cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153 and a tag attached indicating the temperature to which the cooling system can be subjected before damage. The crankcase and all other lubrication reservoirs shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used and its grade and temperature range. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and other servicing required prior to placing the truck in operation.

5.2 Packing (see 6.8). Each complete truck, preserved as specified in 5.1 shall not require packing.

5.3 Marking (see 6.8). Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The truck is intended for use in combating structural-type fires in areas of military operations.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
(c) When truck shall be furnished with a water tower (see 3.21).
(d) Time frame required for submission of the first article model, and number of models required (see 3.2).
(e) When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(f) When initial production inspection is required and number of trucks to be furnished, when applicable (see 3.3).
(g) When altitude modification kits are required (see 3.5.2).
(h) Whether communication equipment or space is required (see 3.18).
(i) When electromagnetic interference requirements shall be other than as specified (see 3.19).

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(j) When lubrication other than as specified is required (see 3.20).
(k) When truck exterior is to be treated and painted other than as specified (see 3.24).
(l) When the welding process will not be at the option of the contractor (see 3.27.4).
(m) When altitude and high temperature pumping test shall be performed on the first article model (see 4.5.2.1.4).
(n) Level of preservation required (see 5.1).

6.3 Definitions. The following definitions shall apply throughout this specification.

6.3.1 Automatic transmission. A transmission in which normal ratio changes are effected automatically without manual assist.

6.3.2 Ambient temperature. The temperature of the atmosphere measured at the intake side of the engine cooling radiator.

6.3.3 Excessive surging. Excessive surging shall be defined as any movement of the water or fuel that would cause the operator of the vehicle to lose control during acceleration, deceleration or any maneuver of the vehicle.

6.4 First article model. Any changes or deviations of production trucks from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article model will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each truck.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of trucks.

6.7 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the truck covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC group 91.

6.8 Marking. Levels of packing have not been included as they are not appropriate for this item. For purposes of preservation/packing level marking, the marking shall be as required by the customer.

6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

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6.10 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, request for proposal, and contracts the following statement:

"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitable finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed."

Custodians:
Army - ME
Navy - MC

Review activities:
Army - CE
DLA - CS

Preparing activity:
Army - ME

Project 4210-0349

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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